

9	(a)	Periodic to and fro motion of a segment of the arrow from its position before release in a direction <u>perpendicular to the motion of arrow/ limb</u> of the bow/ <u>left-right</u>	
	(b)	Energy stored = area under the force-displacement graph	

$$= \frac{1}{2} (350)(0.6)$$

$$= 105 \text{ J}$$

(c)

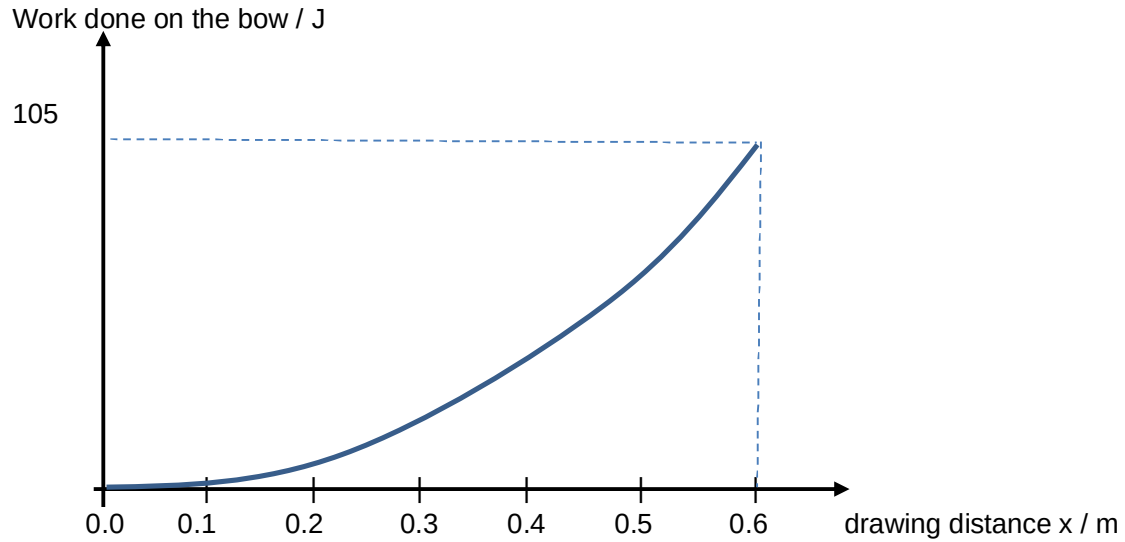
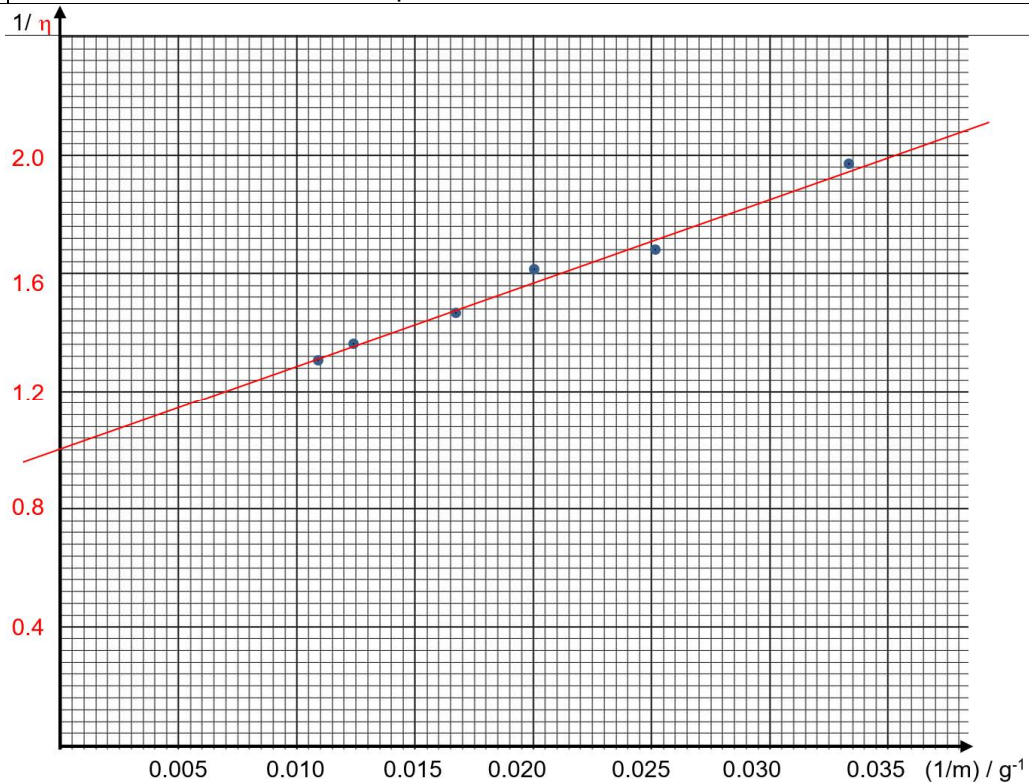


Fig. 9.6

Curve line from 0 J to 105 J when x changes from 0 to 0.6 m
 Zero gradient at $x = 0$ and increasing gradient

(d) (i) Draw the line of best fit for the points



(ii) From $\eta = m / (m + k)$
 $1/\eta = (m + k)/m = 1 + k/m$

If the graph of $1/\eta$ against $1/m$ is a straight line with y-intercept equal to 1

		Since the graph is a straight line with a y-intercept equal to 1, the relation is tested correct	
	(iii)	$k = \text{gradient of the line}$ $= 1.84 - 1.0 / 0.030 - 0 = 28 \text{ g}$ Calculate gradient correctly, read coordinate to $\frac{1}{2}$ smallest grid	
	(iv)	Let $m = 70 \text{ g}$, $1/m = 1/70 = 0.0143$ From the graph $1/\eta = 1.40$ $\eta = 0.714$ $\eta = \text{KE/Energy stored}$ $= \frac{1}{2} m v^2 / E$ $0.714 = \frac{1}{2} (0.070) v^2 / 105$ $v = 46 \text{ m s}^{-1}$	
	(e)	time to clear handle $= (0.15 + 0.60) / 50 = 0.015 \text{ s}$ $1 \frac{1}{4} T = 0.0150 \text{ s}$ $T = 0.012 \text{ s}$ $f = 1/T = 1/0.0120 = 83 \text{ s}^{-1}$	
	(f) (i)		
		Force F / N 500 400 300 200 100 0.1 0.2 0.3 0.4 0.5 0.6 drawing distance x/m Answer to f(i)	
		Straight line from $F = 150 \text{ N}$ to $F = 350 \text{ N}$	
	(ii)	Energy stored $= \frac{1}{2} (350 + 150) \times 0.6 = 150 \text{ N}$ This is greater than that of the Longbow. Hence the arrow can travel at greater speed and further.	

	(g)	As compared to the fully drawn Longbow at same max drawing force (350 N) at the drawing distance of 0.6 N 1. It can store more energy for a fully drawn bow. 2. It allows the archer to hold and aim a fully drawn bow without as much strain or fatigue since the drawing force at the fully drawn distance is smaller.	
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